Lab Session (Open-Ended Lab)

Open Ended Lab	
Bloom's Taxonomy	**GAs**
A3 C3 C5	GA-2 GA-4 GA-6

Title: Designing a Hospital Management System using Object-Oriented Programming Principles

Motivation:

The purpose of this Open-Ended Lab (OEL) is to give students practical experience in applying Object-Oriented Programming (OOP) principles to develop a functional hospital management system. By focusing on real-world scenarios, students will learn to implement key OOP concepts like Encapsulation, Inheritance, Polymorphism, and Abstraction in developing a robust system to handle various hospital operations.

Concept (Problem Statement):

In this lab, students will design and implement a Hospital Management System using Java. This system should manage the relationships between hospitals, patients, doctors, wards, and staff. The system must use OOP principles to create a flexible and efficient solution to handle the interactions between these entities.

Design:

- **Classes and Objects:** Create classes for Hospital, Ward, Person, Patient, Doctor, and Staff. Each class should encapsulate the necessary data and provide methods for interacting with that data. For example, the Doctor class can have methods to manage patient assignments and ward duties.

- **Inheritance:** Implement a hierarchy of people where Patient, Doctor, and Staff inherit from a base Person class. This will allow for the addition of future roles within the hospital system without rewriting core functionality.
- **Polymorphism:** Use polymorphism to handle the relationship between different types of people. For instance, a method to assign a person to a ward can work for both Patient and Doctor by using a common interface.
- **Abstraction:** Create abstract classes or interfaces such as MedicalEntity or Assignable to ensure consistency in assigning people to different hospital facilities. This can include functionality to handle interactions between entities such as WardAssignment or MedicalTreatment.

Key Functionalities:

- **Patient Management:** Implement methods to add, view, and update patient records. These records should track personal details, medical history, current ward, and assigned doctors.
- **Ward Management:** Enable hospitals to manage wards and their capacity. The system should allow for ward assignments based on bed availability and specific medical needs.
- **Doctor and Staff Assignments:** Implement functionality to manage doctor and staff assignments to specific wards or patients. This should include tracking shifts, availability, and specialization.
- **Hospital Reporting:** Generate reports to summarize hospital operations such as patient occupancy per ward, doctor assignments, and patient treatment statuses.

Deliverables

Background/Theory:

Procedure / Methodology:

Flowchart / Block diagram:

Analysis:

Results:

Discussion on Results:

Concluding Remarks:

Reference: